

PG Business Analytics Game

The Airline Analytics Challenge

1. Business Context

Participants act as Business Analytics Consultants supporting an airline's demand-planning team. The airline must estimate how many seats will be sold in Economy and Premium Economy under changing market, pricing, capacity, competition and operational conditions.

The focus is seat demand and the business logic behind the forecast.

2. Objective

Forecast, for each of six live rounds, the number of Economy seats and Premium Economy seats expected to be sold. Each forecast must be accompanied by concise analytical reasoning explaining the main demand drivers.

3. Pre-Game Preparation

- One hour before the game begins, teams receive the historical dataset and game instructions.
- Participants receive a 60-observation monthly historical/simulated training dataset. The historical data contains demand, fare, capacity, competition, promotion, macroeconomic/operational and seasonal indicators together with Economy and Premium Economy seats sold.
- Teams may explore the historical data, engineer features, compare models, calculate coefficients, prepare spreadsheets/code, and develop a forecasting strategy.
- The final six scenarios are not revealed during the preparation period.

4. Live Game Structure

There are six rounds. Each round lasts exactly 10 minutes.

At the beginning of each round, the scenario narrative and all quantitative scenario variables required for the forecasting model are revealed. Participants should translate the business situation into a quantitative demand forecast.

5. Required Submission

- Economy seats forecast (whole number).
- Premium Economy seats forecast (whole number).
- Brief analytical reasoning explaining the forecast, including the most important demand drivers, trade-offs or risks.

6. Scoring

Forecast Accuracy — 60% weightage

Your forecasts will be compared with the actual occupancy observed for the six events.

Accuracy will be calculated separately for:

- Economy occupancy
- Premium Economy occupancy

Both categories will receive balanced consideration.

The competition will consider relative forecasting accuracy so that Premium occupancy is not disadvantaged merely because its absolute numbers are smaller.

Analytical Logic & Strategy — 40% weightage

Your analytical reasoning will account for 40% of the final score.

Judges will consider:

- understanding of the data;
- identification of important demand drivers;
- appropriateness of the analytical method;
- validation and analytical rigor;
- interpretation of the forecasts;
- and the ability to explain the decisions made.

Final Score

Your overall score will be: Maximum **100 points**

The team with the highest final score will be declared the winner.

7. Rules

I. Time Limit

Each round has a strict: **10-minute time limit**

Once the round begins, the countdown starts.

Teams must submit their forecasts before the round closes.

II. No Changes After Submission Deadline

Once a round closes:

- forecasts cannot normally be changed;
- late submissions may not be accepted;
- the organizer's recorded submission will be treated as final.

Teams should therefore avoid waiting until the final seconds to submit.